TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	56.4 ± 6.55
Correlation Stability	75.1 ± 6.98
Distribution Consistency	62.6 ± 6.87
Entropy Preservation	63.2 ± 10.44
Mean Z Score Drift	0.6 ± 0.16
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	0.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	99.8 ± 0.40
Feature Accuracy	57.3 ± 12.08
Novelty Score	0.9 ± 0.03

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Dataset Audit: heart_disease.csv

1. Dataset Portrait

Attribute	Value
Rows	303
Columns	14
Numeric features	14
Categorical features	0
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
age	54.439	54.413	-0.026	0.0%	OK
sex	0.680	0.727	+0.047	7.0%	OK
cp	3.158	3.467	+0.309	9.8%	OK
trestbps	131.690	135.996	+4.306	3.3%	OK
chol	246.693	297.888	+51.195	20.8%	Warn
fbs	0.149	0.124	-0.025	16.5%	Warn
restecg	0.990	1.145	+0.155	15.6%	Warn
thalach	149.607	140.190	-9.417	6.3%	OK
exang	0.327	0.277	-0.050	15.3%	Warn
oldpeak	1.040	2.937	+1.897	182.5%	Drift
slope	1.601	1.917	+0.317	19.8%	Warn
ca	0.672	0.748	+0.076	11.3%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	56.4 ± 6.55
Correlation Stability (*2)	75.1 ± 6.98
Distribution Consistency (*3)	62.6 ± 6.87
Entropy Preservation (*4)	63.2 ± 10.44
Mean Z Score Drift (*5)	0.6 ± 0.16
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	0.0 ± 0.00

Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	99.8 ± 0.40
Feature Accuracy (*10)	57.3 ± 12.08
Novelty Score (*11)	0.9 ± 0.03

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5540983606557377	0.5409836065573771	0.5475409836065573	-0.006557377049180335	
F1	0.4442202071436568 4	0.4183804481651386	0.4434966724342447 3	-0.00072353470941208 81	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5344262295081967	0.5360655737704918	0.5491803278688525	0.01475409836065573 1	
F1	0.5119295254727356	0.4788252408188535	0.5241789366785394	0.01224941120580361 7	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5573770491803279	0.519672131147541	0.5426229508196723	-0.01475409836065572	
F1	0.568660979990255	0.4626850896107946	0.5498930549786436	-0.01876792501161146 3	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5360655737704918	0.4967213114754097 7	0.4868852459016393 4	-0.04918032786885247	
F1	0.5387930265921737	0.4715723574445355	0.4964425226107337 4	-0.04235050398143992	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5475409836065573	0.5426229508196722	0.5442622950819672	-0.00327868852459016 2	
F1	0.3899270600110827	0.4086827764484632	0.3899686853319773 7	4.1625320894705944e-05	

Model: AB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6081967213114754	0.4836065573770491	0.6	-0.00819672131147543	
F1	0.6070236204175147	0.4454591792923261	0.5881680878429711	-0.01885532574543466	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5836065573770491	0.5655737704918032	0.5934426229508196	0.009836065573770503	
F1	0.5305833441817084	0.4800325987104335	0.5431218690083285	0.012538524826620018	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5409836065573771	0.5114754098360657	0.5459016393442623	0.004918032786885224	
F1	0.5301440076096099	0.4665653482177264	0.53167664466319	0.0015326370535801049	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6	0.5557377049180329	0.5754098360655738	-0.024590163934426236	
F1	0.5665897059498607	0.4664312742753547	0.5137037209520285	-0.05288598499783213	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5622950819672134	0.5065573770491804	0.6147540983606556	0.05245901639344264	
F1	0.5403588144737185	0.4386434895353858	0.573243373426234	0.03288455895251552	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
2	67.0	1.0	4.0	160.0	286.0	0.0	2.0	108.0	1.0	1.5
3	67.0	1.0	4.0	120.0	229.0	0.0	2.0	129.0	1.0	2.6
4	37.0	1.0	3.0	130.0	250.0	0.0	0.0	187.0	0.0	3.5
5	41.0	0.0	2.0	130.0	204.0	0.0	2.0	172.0	0.0	1.4
6	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
7	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6
8	57.0	0.0	4.0	120.0	354.0	0.0	0.0	163.0	1.0	0.6
9	63.0	1.0	4.0	130.0	254.0	0.0	2.0	147.0	0.0	1.4
10	53.0	1.0	4.0	140.0	203.0	1.0	2.0	155.0	1.0	3.1
11	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
12	56.0	0.0	2.0	140.0	294.0	0.0	2.0	153.0	0.0	1.3
13	56.0	1.0	3.0	130.0	256.0	1.0	2.0	142.0	1.0	0.6
14	44.0	1.0	2.0	120.0	263.0	0.0	0.0	173.0	0.0	0.0
15	52.0	1.0	3.0	172.0	199.0	1.0	0.0	162.0	0.0	0.5
16	57.0	1.0	3.0	150.0	168.0	0.0	0.0	174.0	0.0	1.6
17	48.0	1.0	2.0	110.0	229.0	0.0	0.0	168.0	0.0	1.0
18	54.0	1.0	4.0	140.0	239.0	0.0	0.0	160.0	0.0	1.2
19	48.0	0.0	3.0	130.0	275.0	0.0	0.0	139.0	0.0	0.2
20	49.0	1.0	2.0	130.0	266.0	0.0	0.0	171.0	0.0	0.6

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	32.0	0.0	3.0	108.0	196.0	0.0	2.0	167.0	0.0	0.0
2	29.0	0.0	2.0	94.0	126.0	0.0	2.0	202.0	0.0	0.0
3	38.0	0.0	3.0	109.0	283.0	0.0	0.0	149.0	0.0	0.1363137353208761
4	55.0	1.0	1.0	136.0	232.0	0.0	2.0	161.0	1.0	5.528530836955024
5	34.0	1.0	2.0	96.0	199.0	0.0	2.0	201.0	0.0	0.0
6	50.0	1.0	3.0	123.0	244.0	0.0	0.0	168.0	0.0	3.524746707161407
7	49.0	0.0	4.0	130.0	269.0	0.0	0.0	163.0	0.0	0.0
8	54.0	1.0	3.0	120.0	258.0	0.0	2.0	147.0	0.0	0.4
9	55.0	1.0	2.0	130.0	262.0	0.0	0.0	155.0	0.0	0.0
10	46.0	1.0	2.0	101.0	197.0	1.0	0.0	156.0	0.0	0.0

B.2 Audit Trace: Fold 2 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	77.0	1.0	4.0	200.0	564.0	0.0	2.0	71.0	1.0	6.2
2	29.0	1.0	2.0	94.0	141.0	0.0	0.0	202.0	0.0	0.0
3	77.0	0.0	3.0	94.0	141.0	0.0	0.0	71.0	0.0	0.0
4	77.0	1.0	4.0	200.0	564.0	0.0	2.0	71.0	0.0	6.2
5	29.0	1.0	2.0	94.0	564.0	0.0	0.0	202.0	0.0	0.0
6	29.0	1.0	2.0	200.0	141.0	0.0	0.0	202.0	0.0	0.0
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	60.0	1.0	4.0	130.0	253.0	0.0	0.0	144.0	1.0	1.4
9	58.0	0.0	2.0	136.0	319.0	1.0	2.0	152.0	0.0	0.0
10	59.0	0.0	4.0	174.0	249.0	0.0	0.0	143.0	1.0	0.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	35.0	1.0	4.0	200.0	564.0	0.0	2.0	71.0	0.0	5.6
2	35.0	1.0	2.0	94.0	564.0	0.0	0.0	195.0	0.0	0.0
3	35.0	1.0	4.0	200.0	564.0	0.0	2.0	71.0	0.0	5.6
4	35.0	1.0	4.0	94.0	564.0	0.0	2.0	71.0	1.0	5.6
5	35.0	1.0	2.0	94.0	126.0	0.0	0.0	195.0	0.0	0.0
6	35.0	1.0	2.0	199.0	564.0	0.0	0.0	195.0	0.0	0.0
7	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
8	45.0	1.0	4.0	115.0	260.0	0.0	2.0	185.0	0.0	0.0
9	45.0	0.0	2.0	130.0	234.0	0.0	2.0	175.0	0.0	0.6
10	46.0	1.0	3.0	150.0	231.0	0.0	0.0	147.0	0.0	3.6

B.4 Audit Trace: Fold 4 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	77.0	1.0	4.0	94.0	564.0	0.0	2.0	71.0	0.0	0.0
2	77.0	1.0	4.0	94.0	564.0	0.0	2.0	71.0	0.0	0.0
3	29.0	1.0	4.0	200.0	126.0	0.0	2.0	202.0	1.0	6.2
4	77.0	1.0	4.0	94.0	564.0	0.0	2.0	71.0	0.0	0.0
5	29.0	1.0	4.0	200.0	126.0	0.0	2.0	202.0	1.0	6.2
6	29.0	1.0	2.0	200.0	126.0	0.0	2.0	202.0	0.0	6.2
7	57.0	1.0	4.0	140.0	192.0	0.0	0.0	148.0	0.0	0.4
8	63.0	0.0	4.0	150.0	407.0	0.0	2.0	154.0	0.0	4.0
9	57.0	0.0	4.0	128.0	303.0	0.0	2.0	159.0	0.0	0.0
10	60.0	1.0	4.0	117.0	230.0	1.0	0.0	160.0	1.0	1.4

B.5 Audit Trace: Fold 5 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	34.0	1.0	4.0	100.0	564.0	0.0	2.0	195.0	0.0	6.2
2	34.0	1.0	4.0	200.0	126.0	0.0	0.0	71.0	1.0	0.0
3	34.0	1.0	4.0	100.0	564.0	0.0	2.0	195.0	0.0	6.2
4	34.0	1.0	2.0	200.0	126.0	0.0	0.0	71.0	1.0	0.0
5	34.0	1.0	2.0	200.0	126.0	0.0	2.0	71.0	0.0	0.0
6	34.0	1.0	2.0	200.0	126.0	0.0	0.0	71.0	0.0	0.0
7	50.0	0.0	2.0	120.0	244.0	0.0	0.0	162.0	0.0	1.1
8	56.0	1.0	2.0	120.0	236.0	0.0	0.0	178.0	0.0	0.8
9	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0
10	64.0	0.0	3.0	140.0	313.0	0.0	0.0	133.0	0.0	0.2

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B.6 Audit Trace: Fold 6 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	77.0	1.0	4.0	200.0	417.0	0.0	2.0	88.0	1.0	6.2
2	77.0	0.0	4.0	200.0	417.0	0.0	2.0	195.0	0.0	0.0
3	34.0	0.0	2.0	200.0	417.0	0.0	0.0	195.0	0.0	0.0
4	34.0	0.0	2.0	200.0	417.0	0.0	2.0	195.0	0.0	0.0
5	77.0	1.0	4.0	200.0	126.0	0.0	2.0	88.0	1.0	6.2
6	66.0	1.0	3.0	94.0	126.0	0.0	0.0	195.0	1.0	6.2
7	39.0	1.0	4.0	118.0	219.0	0.0	0.0	140.0	0.0	1.2
8	62.0	0.0	4.0	138.0	294.0	1.0	0.0	106.0	0.0	1.9
9	66.0	1.0	4.0	160.0	228.0	0.0	2.0	138.0	0.0	2.3
10	39.0	1.0	3.0	140.0	321.0	0.0	2.0	182.0	0.0	0.0

B.7 Audit Trace: Fold 7 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	29.0	1.0	4.0	200.0	564.0	0.0	2.0	71.0	0.0	6.2
2	29.0	1.0	2.0	200.0	564.0	0.0	2.0	202.0	0.0	0.0
3	29.0	1.0	4.0	94.0	564.0	0.0	2.0	71.0	0.0	6.2
4	29.0	1.0	2.0	94.0	126.0	0.0	0.0	202.0	0.0	0.0
5	77.0	1.0	4.0	200.0	564.0	0.0	2.0	202.0	1.0	6.2
6	57.0	1.0	4.0	129.0	564.0	0.0	0.0	71.0	0.0	6.2
7	56.0	0.0	4.0	134.0	409.0	0.0	2.0	150.0	1.0	1.9
8	58.0	1.0	3.0	105.0	240.0	0.0	2.0	154.0	1.0	0.6
9	54.0	1.0	2.0	192.0	283.0	0.0	2.0	195.0	0.0	0.0
10	57.0	1.0	4.0	110.0	201.0	0.0	0.0	126.0	1.0	1.5

B.8 Audit Trace: Fold 8 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	29.0	1.0	4.0	192.0	564.0	0.0	2.0	133.0	1.0	6.2
2	42.0	1.0	2.0	138.0	257.0	0.0	2.0	195.0	1.0	0.0
3	29.0	1.0	4.0	192.0	564.0	0.0	2.0	71.0	1.0	6.2
4	29.0	1.0	4.0	94.0	564.0	0.0	0.0	202.0	0.0	0.0
5	52.0	1.0	4.0	126.0	227.0	0.0	2.0	150.0	1.0	4.321488836440105
6	77.0	1.0	4.0	192.0	564.0	0.0	2.0	202.0	1.0	0.0
7	52.0	1.0	1.0	118.0	186.0	0.0	2.0	190.0	0.0	0.0
8	54.0	0.0	3.0	135.0	304.0	1.0	0.0	170.0	0.0	0.0
9	61.0	1.0	4.0	140.0	207.0	0.0	2.0	138.0	1.0	1.9
10	61.0	1.0	3.0	150.0	243.0	1.0	0.0	137.0	1.0	1.0

B.9 Audit Trace: Fold 9 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	77.0	1.0	4.0	192.0	131.0	0.0	1.0	71.0	0.0	6.2
2	77.0	1.0	3.0	192.0	217.0	0.0	2.0	202.0	0.0	0.0
3	77.0	0.0	4.0	192.0	131.0	0.0	1.0	71.0	0.0	6.2
4	29.0	1.0	2.0	94.0	131.0	0.0	2.0	202.0	0.0	0.0
5	29.0	0.0	4.0	192.0	131.0	0.0	1.0	71.0	0.0	6.2
6	77.0	0.0	4.0	192.0	131.0	0.0	1.0	71.0	0.0	6.2
7	44.0	0.0	3.0	118.0	242.0	0.0	0.0	149.0	0.0	0.3
8	38.0	1.0	1.0	120.0	231.0	0.0	0.0	182.0	1.0	3.8
9	49.0	1.0	3.0	120.0	188.0	0.0	0.0	139.0	0.0	2.0
10	61.0	1.0	4.0	120.0	260.0	0.0	0.0	140.0	1.0	3.6

B.10 Audit Trace: Fold 10 | Gen: 100%

#	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak
1	77.0	1.0	4.0	200.0	564.0	0.0	0.0	71.0	1.0	5.6
2	29.0	0.0	2.0	200.0	564.0	0.0	2.0	202.0	0.0	0.0
3	77.0	1.0	4.0	94.0	564.0	0.0	0.0	71.0	1.0	5.6
4	29.0	1.0	2.0	200.0	126.0	0.0	2.0	202.0	0.0	0.0
5	77.0	1.0	4.0	200.0	564.0	0.0	0.0	71.0	1.0	5.6
6	29.0	1.0	2.0	94.0	126.0	0.0	2.0	202.0	0.0	0.0
7	63.0	0.0	3.0	135.0	252.0	0.0	2.0	172.0	0.0	0.0
8	63.0	1.0	1.0	145.0	233.0	1.0	2.0	150.0	0.0	2.3
9	60.0	1.0	4.0	125.0	258.0	0.0	2.0	141.0	1.0	2.8
10	62.0	0.0	4.0	140.0	268.0	0.0	2.0	160.0	0.0	3.6